Figure 15-4 shows the frequency response of two other relatives of the moving average filter. When a pure **Gaussian** is used as a filter kernel, the frequency response is also a Gaussian, as discussed in Chapter 11. The Gaussian is important because it is the impulse response of many natural and manmade systems. For example, a brief pulse of light entering a long fiber optic transmission line will exit as a Gaussian pulse, due to the different paths taken by the photons within the fiber. The Gaussian filter kernel is also used extensively in *image processing* because it has unique properties that allow fast two-dimensional convolutions (see Chapter 24). The second frequency response in Fig. 15-4 corresponds to using a **Blackman window** as a filter kernel. (The term *window* has no meaning here; it is simply part of the accepted name of this curve). The exact shape of the Blackman window is given in Chapter 16 (Eq. 16-2, Fig. 16-2); however, it looks much like a Gaussian.

How are these relatives of the moving average filter better than the moving average filter itself? Three ways: First, and most important, these filters have better *stopband attenuation* than the moving average filter. Second, the filter kernels *taper* to a smaller amplitude near the ends. Recall that each point in the output signal is a weighted sum of a group of samples from the input. If the filter kernel tapers, samples in the input signal that are farther away are given less weight than those close by. Third, the step responses are *smooth* curves, rather than the abrupt straight line of the moving average. These last two are usually of limited benefit, although you might find applications where they are genuine advantages.

The moving average filter and its relatives are all about the same at reducing random noise while maintaining a sharp step response. The ambiguity lies in how the *risetime* of the step response is measured. If the risetime is measured from 0% to 100% of the step, the moving average filter is the best you can do, as previously shown. In comparison, measuring the risetime from 10% to 90% makes the Blackman window *better* than the moving average filter. The point is, this is just theoretical squabbling; consider these filters equal in this parameter.

The biggest difference in these filters is *execution speed*. Using a recursive algorithm (described next), the moving average filter will run like lightning in your computer. In fact, it is the *fastest* digital filter available. Multiple passes of the moving average will be correspondingly slower, but still very quick. In comparison, the Gaussian and Blackman filters are excruciatingly slow, because they must use convolution. Think a factor of ten times the number of points in the filter kernel (based on multiplication being about 10 times slower than addition). For example, expect a 100 point Gaussian to be 1000 times slower than a moving average using recursion.

Recursive Implementation

A tremendous advantage of the moving average filter is that it can be implemented with an algorithm that is very fast. To understand this